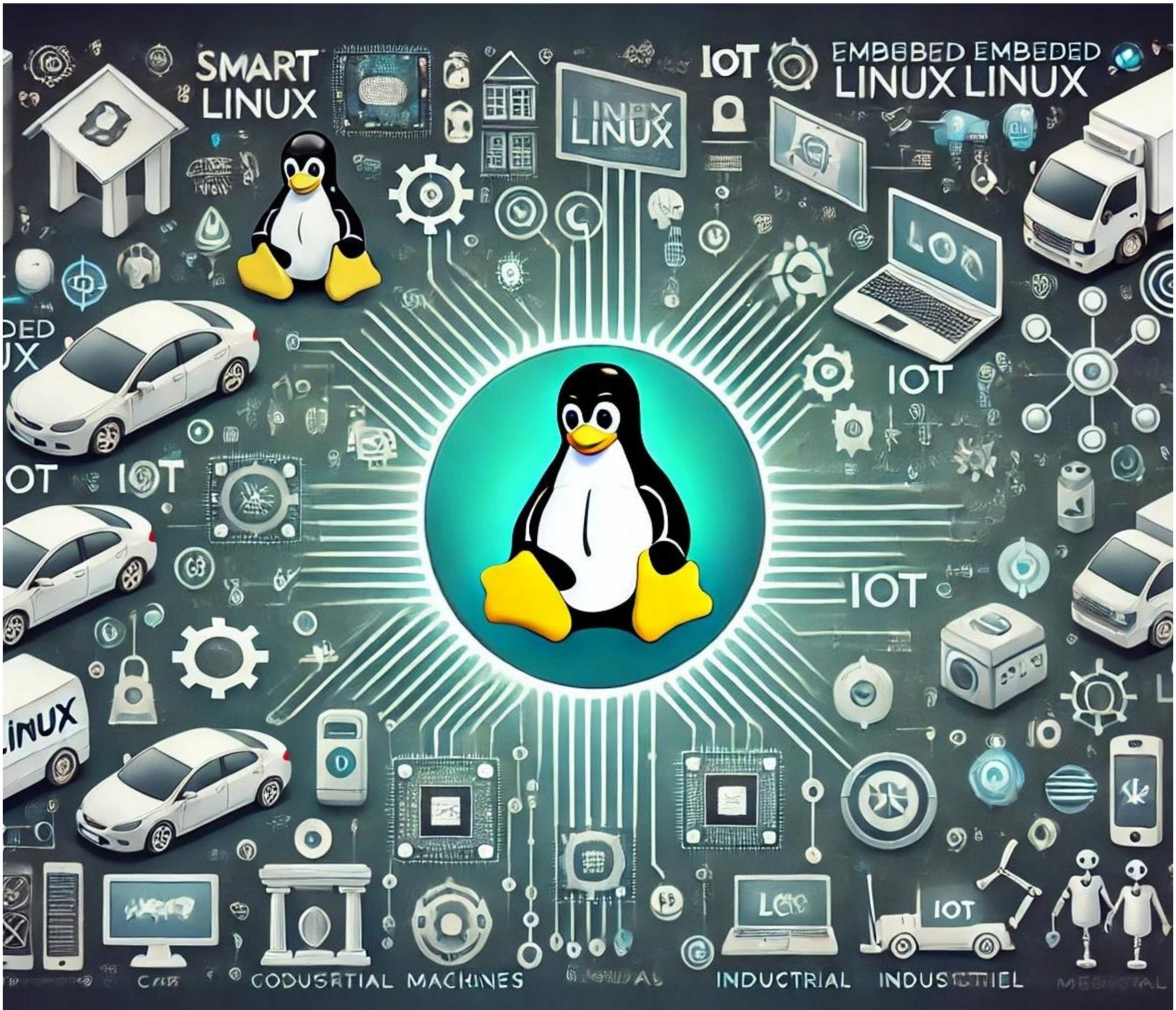

1400+ Embedded Linux Interview Questions

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Learning Series from scratch to expert

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200 Beginner-Level Embedded Linux Interview Questions

System Setup and Configuration (40 Questions)

1. What is embedded Linux, and how does it differ from desktop Linux?
2. Name three popular Linux distributions used in embedded systems.
3. How do you install Raspbian on a Raspberry Pi using an SD card?
4. What is the purpose of the apt-get update command?
5. How do you upgrade all packages on a Debian-based embedded system?
6. Explain how to set the system timezone using timedatectl.
7. How do you change the hostname of an embedded Linux device?
8. What steps are required to create a new user with sudo privileges?
9. How do you enable SSH access on a Raspberry Pi?
10. How can you disable the GUI to boot into the command-line interface?
11. What is the purpose of the systemctl command in embedded Linux?
12. How do you install a specific version of a package using apt-get?
13. How do you check available disk space on an embedded Linux system?
14. What is the process to mount a USB drive on an embedded device?
15. How do you safely unmount a USB drive to prevent data loss?
16. How can you check the kernel version running on your device?
17. What steps are needed to install the nano text editor?
18. How do you configure automatic updates for security patches?
19. What is a swap file, and how do you create one on an embedded device?
20. How do you check system uptime using a command?
21. How do you reboot an embedded Linux device using systemctl?
22. What command shuts down an embedded Linux system gracefully?
23. How do you display CPU information on an embedded device?
24. What is the purpose of the /etc directory in Linux?
25. How do you install a lightweight desktop environment like LXDE?
26. How can you change the default shell to bash on an embedded system?
27. What steps are needed to back up the /etc directory to a USB drive?
28. How do you restore a backed-up /etc directory?
29. How do you configure a static DNS server on an embedded device?
30. How do you check available memory using the free command?
31. What is a cron job, and how do you schedule one to run daily?
32. How do you check the version of the bootloader (e.g., U-Boot)?
33. What is the purpose of the raspi-config tool on Raspberry Pi?
34. How do you enable the root user account on an embedded Linux system?
35. How do you configure a device to boot into single-user mode?
36. What command displays the current runlevel or target in systemd?
37. How do you update the firmware on a Raspberry Pi?
38. What is the purpose of the /boot directory in embedded Linux?
39. How do you check the status of a specific service using systemctl?
40. How do you configure a device to use a specific locale?

File System and Storage (30 Questions)

41. What is the purpose of the /home directory in Linux?

42. How do you create a new directory using the command line?
43. What command copies a file from one directory to another?
44. How do you move a file to a different directory?
45. What is the command to delete a file in Linux?
46. How do you create an empty file using the touch command?
47. How do you list all files, including hidden files, in a directory?
48. What command displays the permissions of a file?
49. How do you change file permissions using chmod?
50. How do you change file ownership using chown?
51. What is a symbolic link, and how do you create one?
52. How do you find a file by name using the find command?
53. What is the purpose of the grep command in file operations?
54. How do you compress a directory into a .tar.gz file?
55. How do you extract a .tar.gz file to a specific directory?
56. What command checks the size of a directory?
57. How do you create a read-only file using permissions?
58. What is the difference between >> and > when redirecting output?
59. How do you display the contents of a file using cat?
60. How do you view the first 10 lines of a file?
61. What is the purpose of the /proc filesystem?
62. How do you check the type of a filesystem on a partition?
63. What command displays the last 10 lines of a file?
64. How do you create a hard link to a file?
65. What is the purpose of the du command in Linux?
66. How do you search for a string across multiple files using grep?
67. How do you delete a directory and all its contents?
68. What is the purpose of the mount command in Linux?
69. How do you check if a filesystem is mounted correctly?
70. How do you append text to a file without overwriting it?

Networking (30 Questions)

71. How do you check the IP address of an embedded Linux device?
72. What is the purpose of the /etc/network/interfaces file?
73. How do you configure a static IP address on an embedded device?
74. How do you restart networking services after a configuration change?
75. What is the ping command used for in networking?
76. How do you use curl to download a file from a URL?
77. What command checks for open ports on a device?
78. How do you install and use nmap to scan a network?
79. How do you set up a simple HTTP server using Python?
80. How do you connect to a Wi-Fi network using nmcli?
81. What is the difference between ifconfig and ip addr?
82. How do you enable or disable a network interface?
83. What is the purpose of the nslookup command?
84. How do you install and configure an FTP server like vsftpd?
85. How do you transfer a file to another device using scp?
86. What steps are needed to set up passwordless SSH login?

87. How do you monitor network traffic using iftop?
88. How do you change the default SSH port on a device?
89. What command displays the routing table on a Linux system?
90. How do you enable port forwarding using iptables?
91. What is a DHCP client, and how do you configure it?
92. How do you check the status of a network interface?
93. What is the purpose of the netstat command?
94. How do you test DNS resolution on an embedded device?
95. How do you configure a device to act as a DHCP client?
96. What is the purpose of the /etc/hosts file?
97. How do you check for network packet loss using ping?
98. What is the traceroute command used for?
99. How do you install and use wget to download files?
100. How do you check the MAC address of a network interface?

Shell Scripting and Programming (30 Questions)

101. What is a shell script, and why is it useful in embedded Linux?
102. How do you write a shell script to print "Hello, World!"?
103. How do you make a shell script executable?
104. What is the purpose of the shebang line (#!/bin/bash) in a script?
105. How do you list all files in a directory using a shell script?
106. How do you check if a file exists in a shell script?
107. How do you write a shell script to display the current date?
108. How do you write a shell script to calculate the sum of two numbers?
109. What is the purpose of the if statement in a shell script?
110. How do you write a shell script to back up a directory?
111. How do you check disk usage in a shell script?
112. How do you count the number of lines in a file using a script?
113. How do you write a shell script to rename files with a specific extension?
114. How do you check if a process is running in a shell script?
115. What is the purpose of the for loop in a shell script?
116. How do you write a Python script to read a file's contents?
117. How do you compile a simple C program on an embedded device?
118. How do you write a Python script to monitor CPU temperature?
119. What is the purpose of the chmod +x command for scripts?
120. How do you write a shell script to automate package updates?
121. How do you log system uptime to a file using a script?
122. How do you write a Python script to ping a server?
123. How do you check available memory in a Python script?
124. How do you write a shell script to restart a service?
125. What is the purpose of the while loop in a shell script?
126. How do you write a Python script to control a GPIO pin?
127. How do you write a shell script to check for a USB device?
128. How do you append a timestamp to a log file in a script?
129. How do you write a Python script to fetch weather data using an API?
130. How do you write a shell script to monitor a directory for changes?

Kernel and Boot Process (30 Questions)

131. What is the Linux kernel, and what is its role in embedded systems?
132. How do you check the kernel boot parameters in /proc/cmdline?
133. What is the purpose of the GRUB bootloader in embedded Linux?
134. How do you update the GRUB configuration on an embedded device?
135. How do you check loaded kernel modules using lsmod?
136. What is the purpose of the modprobe command?
137. How do you unload a kernel module using rmmod?
138. How do you view kernel logs using dmesg?
139. What is the role of the initramfs in the boot process?
140. How do you configure the kernel to boot in verbose mode?
141. What is the difference between U-Boot and GRUB bootloaders?
142. How do you modify the boot delay in U-Boot?
143. How do you check the kernel version and build date?
144. What is the purpose of the /boot directory in the boot process?
145. How do you enable a kernel module to load at boot?
146. How do you check for kernel errors in the ring buffer?
147. What is the difference between systemd and SysVinit?
148. How do you create a simple systemd service to run at boot?
149. How do you check boot time using systemd-analyze?
150. How do you disable a kernel module from loading at boot?
151. What is the purpose of the device tree in embedded Linux?
152. How do you check the kernel command line arguments?
153. What is the role of the init process in Linux?
154. How do you configure a device to boot from a specific partition?
155. What is the purpose of the zImage file in the boot process?
156. How do you back up the current kernel and initramfs?
157. What is the difference between a kernel module and a built-in driver?
158. How do you check the status of the init system?
159. What is the purpose of the /proc filesystem in kernel operations?
160. How do you configure a custom kernel boot logo?

Hardware Interaction and GPIO (30 Questions)

161. What is GPIO, and how is it used in embedded Linux?
162. How do you enable I2C on a Raspberry Pi using raspi-config?
163. How do you read the status of a GPIO pin using /sys/class/gpio?
164. How do you write a script to toggle a GPIO pin on and off?
165. How do you connect an LED to a GPIO pin and control it?
166. What steps are needed to read input from a push button via GPIO?
167. What is the purpose of the i2c-tools package?
168. How do you enable SPI on an embedded Linux device?
169. How do you read data from a temperature sensor via I2C?
170. What is PWM, and how is it used in embedded systems?
171. How do you write a Python script to control a servo motor via PWM?
172. How do you connect a relay module to a GPIO pin?
173. What is an ADC, and how is it used with embedded Linux?

174. How do you configure a GPIO pin as an interrupt source?
175. How do you write a script to monitor a GPIO pin for changes?
176. How do you control a buzzer connected to a GPIO pin?
177. What steps are needed to interface a 16x2 LCD display?
178. How do you read data from a DHT11 humidity sensor?
179. How do you configure a GPIO pin to control a fan?
180. How do you write a Python script to log sensor data?
181. What is the purpose of the /sys/class/gpio directory?
182. How do you interface a motion sensor (PIR) with a GPIO pin?
183. What is the difference between I2C and SPI protocols?
184. How do you check if an I2C device is detected on the bus?
185. How do you configure a GPIO pin for input with a pull-up resistor?
186. What is the purpose of the dtparam command in config.txt?
187. How do you interface a simple analog sensor using an ADC?
188. How do you control the brightness of an LED using PWM?
189. What is the role of the device tree in GPIO configuration?
190. How do you troubleshoot a GPIO pin that is not responding?

Debugging and Troubleshooting (40 Questions)

191. What is the purpose of the journalctl command in debugging?
192. How do you find the process ID of a running service?
193. How do you kill a process using the kill command?
194. What is the purpose of the /var/log/syslog file?
195. How do you monitor system processes using top?
196. What is the difference between top and htop?
197. How do you check for failed systemd services?
198. What steps would you take to debug a failed SSH connection?
199. How do you troubleshoot a network connectivity issue?
200. How do you check for disk errors using fsck?

Notes

- **Target Audience**: These questions are designed for beginners applying for entry-level roles in embedded Linux development, such as junior engineers or interns working with SBCs like Raspberry Pi.
- **Preparation Tips**: Candidates should practice hands-on tasks (e.g., setting up a Raspberry Pi, writing simple scripts, configuring GPIO) and review Linux basics (e.g., file systems, networking, kernel concepts).
- **Practical Focus**: Many questions test practical skills, so hands-on experience with an embedded Linux device is essential.
- **Resources**: Refer to the Linux man pages, Raspberry Pi documentation, and online tutorials (e.g., elinux.org) for deeper understanding.

300 Intermediate-Level Embedded Linux Interview Questions

System Administration and Configuration (50 Questions)

1. What is the difference between a minimal and full root filesystem in embedded Linux?
2. How do you configure a read-only root filesystem to prevent corruption?
3. What is the purpose of logrotate, and how do you configure it?
4. How do you set up a watchdog timer using systemd?
5. What steps are required to sync system time with an NTP server?
6. How do you configure tmpfs for temporary file storage?
7. How do you optimize boot time by disabling unnecessary services?
8. What is the process to back up a filesystem to a remote server?
9. How do you modify kernel command-line parameters at boot?
10. What is a chroot environment, and how do you set one up?
11. How do you configure rsyslog to forward logs to a remote server?
12. What is an initramfs, and how do you create a custom one?
13. How do you set up a RAID1 array using two USB drives?
14. What steps are needed to boot from an NFS-mounted root filesystem?
15. How do you use apparmor to restrict an application's access?
16. What is plymouth, and how do you configure a custom splash screen?
17. How do you create a custom udev rule to rename a network interface?
18. What is network bonding, and how do you configure it?
19. How do you set up a local package repository using apt-ftparchive?
20. What are cgroups, and how do you limit CPU usage for a process?
21. How do you configure a custom kernel panic handler?
22. What is systemd-networkd, and how do you configure it?
23. How do you implement secure boot using U-Boot?
24. What steps are needed to install a specific kernel version from a custom repository?
25. How do you configure zram for compressed swap space?
26. What is the purpose of systemd-tmpfiles?
27. How do you boot a device in single-user mode?
28. What steps are required to configure a specific locale and keyboard layout?
29. How do you create a systemd timer for weekly backups?
30. What is dnsmasq, and how do you configure it as a DNS cache?
31. How do you configure auditd to monitor file access?
32. What is systemd-nspawn, and how do you use it for containers?
33. How do you configure a CPU power-saving mode?
34. What is the purpose of /etc/fstab, and how do you customize it?
35. How do you automate service restarts using a script?
36. What is a kernel module parameter, and how do you set it at boot?
37. How do you install and configure snapd for snap packages?
38. How do you configure a custom /etc/hosts file for DNS?
39. What steps are needed to rotate and compress logs older than 7 days?
40. How do you configure selinux in permissive mode?
41. What is the difference between systemd and runit?
42. How do you configure a custom boot logo in the kernel?

43. What is the purpose of systemd-analyze, and how do you use it?
44. How do you configure a device to use a specific time zone for logs?
45. What is the role of /etc/sysctl.conf in system configuration?
46. How do you set up a custom systemd service with dependencies?
47. What is the purpose of update-rc.d in SysVinit systems?
48. How do you configure a device to use a specific GRUB timeout?
49. What is the difference between apt and dpkg?
50. How do you configure a device to use a custom kernel command-line?

Kernel and Device Drivers (50 Questions)

51. What is the process to compile a Linux kernel for an ARM device?
52. How do you enable a specific kernel configuration using menuconfig?
53. What is cross-compilation, and how do you set it up for a kernel module?
54. What is a character device driver, and what are its key components?
55. How do you apply the PREEMPT_RT patch to a kernel?
56. What is the purpose of a kernel patch, and how do you apply one?
57. How do you enable USB device driver support in the kernel?
58. What is the role of sysfs in kernel modules?
59. How do you use kconfig to configure kernel options?
60. What is a device tree, and how do you modify it?
61. How do you debug a kernel oops using dmesg?
62. What is the purpose of kernel debugging symbols?
63. How do you configure the kernel to support an SPI device driver?
64. What is a kernel module, and how do you write a simple one?
65. How do you use kprobes to trace a kernel function?
66. What steps are needed to configure I2C device driver support?
67. How do you handle interrupts in a kernel module?
68. What is the purpose of a device tree overlay?
69. How do you configure the kernel for a specific network driver?
70. What is the role of printk in kernel debugging?
71. How do you enable a specific filesystem in the kernel?
72. What is the purpose of power management in the kernel?
73. How do you use ftrace to trace kernel events?
74. What is a platform driver, and how does it differ from a character driver?
75. How do you configure the kernel for a specific audio codec?
76. What is the purpose of devicetree bindings?
77. How do you debug a kernel module using gdb?
78. What is the role of modprobe in kernel module management?
79. How do you configure the kernel for a USB gadget driver?
80. What is the difference between a kernel module and a built-in driver?
81. How do you check kernel module dependencies?
82. What is the purpose of the initramfs in the boot process?
83. How do you configure the kernel to support a specific GPU driver?
84. What is the role of kallsyms in kernel debugging?
85. How do you enable a kernel module to load at boot?
86. What is the purpose of the dtb file in embedded Linux?
87. How do you configure the kernel for a specific CAN bus driver?

88. What is the role of irqchip in interrupt handling?
89. How do you write a kernel module to expose a sysfs entry?
90. What is the purpose of kbuild in kernel compilation?
91. How do you configure the kernel for a specific touch controller?
92. What is the role of make modules_install in kernel compilation?
93. How do you debug a kernel panic using logs?
94. What is the purpose of the zImage versus Image in kernel builds?
95. How do you configure the kernel for a specific PCIe driver?
96. What is the role of devm_ functions in kernel programming?
97. How do you use kmod tools to manage kernel modules?
98. What is the purpose of CONFIG_LOCALVERSION in kernel configuration?
99. How do you configure the kernel for a specific wireless driver?
100. What is the role of kobject in kernel driver development?

Networking and Communication (50 Questions)

101. How do you configure a device as a Wi-Fi access point using hostapd?
102. What is the purpose of openvpn, and how do you set it up?
103. How do you configure VLAN tagging on a network interface?
104. What is the difference between a TCP server and a UDP server?
105. How do you enable IPv6 addressing on an embedded device?
106. What is a DHCP server, and how do you configure one?
107. How do you monitor network bandwidth using vnstat?
108. What is a reverse SSH tunnel, and how do you set it up?
109. How do you block incoming traffic except SSH using iptables?
110. What is the purpose of a UDP client, and how do you implement one?
111. How do you configure a specific MTU size for a network interface?
112. What is a network bridge, and how do you set one up?
113. How do you log dropped network packets?
114. What steps are needed to configure a static IPv6 address?
115. How do you set up mosquitto for MQTT communication?
116. What is the purpose of the ip command in networking?
117. How do you configure wpa_supplicant for enterprise Wi-Fi?
118. What is the purpose of iwlist in Wi-Fi management?
119. How do you configure Quality of Service (QoS) for network traffic?
120. What is samba, and how do you configure it for file sharing?
121. How do you monitor network latency using mtr?
122. What is the purpose of an NTP server, and how do you configure one?
123. How do you use avahi for mDNS service discovery?
124. What is a WebSocket, and how do you implement one in Python?
125. How do you configure a proxy server for HTTP traffic?
126. What is openconnect, and how do you use it for VPN?
127. How do you log incoming SSH attempts?
128. What is multicast, and how do you configure a device to use it?
129. How do you configure a device as a DHCP client?
130. What is the purpose of ethtool in network configuration?
131. How do you set up a device to use a specific DNS resolver?
132. What is the difference between ping and traceroute?

133. How do you configure a device to use a specific network gateway?
134. What is the purpose of nmap in network troubleshooting?
135. How do you set up a simple FTP server using vsftpd?
136. What is the difference between a public and private IP address?
137. How do you configure a device to use a specific subnet mask?
138. What is the purpose of tcpdump in network analysis?
139. How do you set up a device to use a specific VLAN ID?
140. What is the role of ifconfig versus ip commands?
141. How do you configure a device to use a specific SSID for Wi-Fi?
142. What is the purpose of iwconfig in wireless networking?
143. How do you check for open ports using ss?
144. What is the difference between http and https?
145. How do you configure a device to use a specific network interface priority?
146. What is the purpose of route in network configuration?
147. How do you set up a device to use a specific firewall rule?
148. What is the difference between NAT and PAT in networking?
149. How do you configure a device to use a specific MAC address?
150. What is the purpose of dig in DNS troubleshooting?

Shell Scripting and Programming (50 Questions)

151. How do you write a shell script to monitor CPU usage?
152. What is the purpose of a Python virtual environment in embedded Linux?
153. How do you write a Python script to interface with an SPI device?
154. What is the role of sigaction in a C program?
155. How do you write a Python script to log sensor data to a SQLite database?
156. What is the purpose of make in compiling C programs?
157. How do you automate kernel module loading with a script?
158. What is the python-can library, and how do you use it?
159. How do you write a C program to read from a UART device?
160. What is the purpose of OpenCV in embedded Linux?
161. How do you write a script to monitor memory usage?
162. What is the difference between a shell script and a Python script?
163. How do you write a Python script to control a relay via MQTT?
164. What is the purpose of cron in scheduling scripts?
165. How do you write a script to automate firmware updates?
166. What is the purpose of json in Python scripting?
167. How do you write a C program to handle an I2C protocol?
168. What is the purpose of popen in a C program?
169. How do you write a Python script to log GPS data?
170. What is the difference between fork and exec in C programming?
171. How do you write a script to monitor a network service?
172. What is the pybluez library, and how do you use it?
173. How do you write a C program to handle a GPIO interrupt?
174. What is the purpose of subprocess in Python?
175. How do you write a Python script to control a DC motor?
176. What is the role of scp in script automation?
177. How do you write a Python script to interface with a LoRa module?

178. What is the purpose of signal handling in C programs?
179. How do you write a script to log system temperature?
180. What is the difference between a compiled and interpreted language?
181. How do you write a Python script to fetch data from a REST API?
182. What is the purpose of pthread in C programming?
183. How do you write a script to copy files based on modification time?
184. What is the role of struct in C programming?
185. How do you write a Python script to log sensor data to a CSV file?
186. What is the purpose of os module in Python?
187. How do you write a C program to interface with a 1-Wire sensor?
188. What is the difference between static and dynamic libraries?
189. How do you write a script to email a log file?
190. What is the purpose of argparse in Python scripting?
191. How do you write a Python script to control a touchscreen?
192. What is the role of stdio.h in C programming?
193. How do you write a script to check for USB device connections?
194. What is the purpose of logging module in Python?
195. How do you write a C program to handle file I/O?
196. What is the difference between make and cmake?
197. How do you write a Python script to monitor a directory for changes?
198. What is the purpose of fcntl.h in C programming?
199. How do you write a script to automate system diagnostics?
200. What is the role of sys module in Python?

Hardware Interfacing and Peripherals (40 Questions)

201. What is the difference between GPIO and PWM in hardware interfacing?
202. How do you configure SPI on an embedded Linux device?
203. What is the purpose of an I2C multiplexer?
204. How do you read data from a barometric pressure sensor via I2C?
205. What is the role of a device tree in hardware interfacing?
206. How do you interface a gyroscope sensor using SPI?
207. What is the purpose of a UART in embedded systems?
208. How do you control a matrix keypad using GPIO pins?
209. What is the difference between an ADC and a DAC?
210. How do you interface an OLED display using I2C?
211. What is the purpose of spidev in Linux?
212. How do you control a stepper motor using GPIO?
213. What is the role of a proximity sensor in embedded systems?
214. How do you configure a GPIO pin as an interrupt source?
215. What is the purpose of a relay module in hardware control?
216. How do you read data from a GPS module using UART?
217. What is the difference between I2C and UART protocols?
218. How do you control a buzzer with variable frequencies?
219. What is the purpose of a device tree overlay for SPI devices?
220. How do you interface a thermal camera module?
221. What is the role of i2c-tools in hardware debugging?
222. How do you control a servo motor using PWM?

223. What is the purpose of a capacitive touch sensor?
224. How do you read data from a humidity sensor?
225. What is the difference between a pull-up and pull-down resistor?
226. How do you interface a 16x2 LCD display with GPIO?
227. What is the purpose of libgpiod in GPIO programming?
228. How do you control a fan using a GPIO pin?
229. What is the role of a motion sensor (PIR) in embedded systems?
230. How do you configure a device to use a specific I2C bus?
231. What is the purpose of a rotary encoder, and how do you interface it?
232. How do you read analog input using an ADC like MCP3008?
233. What is the difference between SPI and I2S protocols?
234. How do you control an LED's brightness using PWM?
235. What is the purpose of a temperature sensor in embedded systems?
236. How do you interface a color sensor to read RGB values?
237. What is the role of dtc in device tree compilation?
238. How do you configure a device to use a specific SPI clock speed?
239. What is the purpose of a pressure sensor in embedded applications?
240. How do you interface a microphone module for audio recording?

Real-Time Systems and Performance (30 Questions)

241. What is real-time scheduling, and how does it apply to embedded Linux?
242. How do you configure a task to use SCHED_FIFO?
243. What is the purpose of the PREEMPT_RT patch?
244. How do you measure interrupt latency in a Linux system?
245. What is the role of chrt in real-time scheduling?
246. How do you optimize a system for low-latency performance?
247. What is cyclictest, and how do you use it?
248. How do you configure a device to use a specific CPU core?
249. What is the purpose of isolcpus in real-time systems?
250. How do you implement a real-time data logger?
251. What is the difference between soft and hard real-time systems?
252. How do you measure system jitter in a real-time application?
253. What is the purpose of nice in process prioritization?
254. How do you configure a device to use cpufreq for performance?
255. What is the role of SCHED_RR in real-time scheduling?
256. How do you optimize a Python script for real-time processing?
257. What is the purpose of taskset in CPU affinity?
258. How do you measure boot time performance?
259. What is the role of irqaffinity in real-time systems?
260. How do you implement a real-time control loop in Python?
261. What is the difference between PREEMPT and PREEMPT_RT?
262. How do you configure a device for low-latency audio?
263. What is the purpose of valgrind in performance optimization?
264. How do you implement a real-time sensor polling script?
265. What is the role of cpuset in resource isolation?
266. How do you measure context switch latency?
267. What is the purpose of sched_setaffinity in C programming?

268. How do you optimize a system for minimal power consumption?
269. What is the role of rtirq in interrupt prioritization?
270. How do you implement a real-time watchdog in a script?

Debugging and Troubleshooting (30 Questions)

271. How do you use strace to debug an application?
272. What is the purpose of journalctl in debugging?
273. How do you identify a memory leak in a C program?
274. What is the role of gdb in debugging applications?
275. How do you analyze network packet loss using tcpdump?
276. What is the purpose of systemctl --failed in troubleshooting?
277. How do you use lsof to find open files by a process?
278. What is the role of kdump in kernel debugging?
279. How do you use perf to analyze system performance?
280. What steps are needed to debug a network connectivity issue?
281. How do you check for errors in /var/log/messages?
282. What is the purpose of sar in performance monitoring?
283. How do you debug a Python script using pdb?
284. What is the role of htop in process troubleshooting?
285. How do you use dmesg to troubleshoot hardware issues?
286. What is the purpose of wireshark in network debugging?
287. How do you debug a GPIO interrupt issue?
288. What is the role of slabtop in memory analysis?
289. How do you troubleshoot a failing network interface?
290. What is the purpose of systemtap in debugging?
291. How do you debug a failed systemd service?
292. What is the role of top versus htop in monitoring?
293. How do you use i2cdump to debug I2C communication?
294. What steps are needed to troubleshoot a USB device failure?
295. How do you analyze CPU usage spikes?
296. What is the purpose of fsck in filesystem troubleshooting?
297. How do you debug a boot failure using logs?
298. What is the role of ethtool in network troubleshooting?
299. How do you use audit2allow to debug SELinux issues?
300. What is the purpose of dmesg -w in real-time debugging?

Notes

- These questions target intermediate-level candidates with practical experience in embedded Linux, assuming familiarity with basic commands, file systems, and hardware interfacing.
- Candidates should be comfortable with kernel configuration, basic driver concepts, networking protocols, real-time scheduling, and debugging tools.
- Practical preparation involves hands-on experience with embedded platforms (e.g., Raspberry Pi), cross-compilation, and tools like gcc, gdb, and systemd.

400 Advanced-Level Embedded Linux Interview Questions

Kernel Development and Optimization (80 Questions)

1. How do you optimize the Linux kernel for a low-latency real-time system?
2. What steps are required to cross-compile a kernel for a custom RISC-V SoC?
3. How do you apply a custom kernel patch to support a proprietary hardware peripheral?
4. What is the role of CONFIG_PREEMPT_RT in achieving deterministic scheduling?
5. How do you configure the kernel to use cgroups v2 for resource isolation?
6. What is the process to implement a custom kernel scheduler for a specific workload?
7. How do you enable kernel lockdown mode with custom integrity policies?
8. What is the purpose of kexec, and how do you configure it for fast reboot?
9. How do you optimize the kernel for minimal boot time (<100ms)?
10. What is the role of transparent huge pages in kernel memory management?
11. How do you write a kernel module to manage a custom DMA engine?
12. What steps are needed to configure the kernel for a specific PCIe root complex?
13. How do you use ftrace to profile kernel performance under heavy load?
14. What is the process to implement a custom interrupt controller driver?
15. How do you configure the kernel to support a custom MIPI CSI-2 camera?
16. What is the role of kprobes in dynamic kernel tracing?
17. How do you optimize the kernel for minimal power consumption in a battery-powered device?
18. What steps are required to support a custom cryptographic hardware accelerator?
19. How do you implement a device tree overlay for dynamic peripheral loading?
20. What is the purpose of kcov in kernel code coverage analysis?
21. How do you configure the kernel to support a specific GPU driver with power management?
22. What is the process to handle a custom clock framework in the kernel?
23. How do you use perf to identify kernel performance bottlenecks?
24. What steps are needed to configure the kernel for a specific Ethernet PHY driver?
25. How do you implement a kernel module to manage a custom thermal zone?
26. What is the role of kunit in kernel module testing?
27. How do you configure the kernel for a specific USB Type-C controller?
28. What is the process to implement a custom SPI controller driver?
29. How do you enable kernel support for a specific audio codec driver?
30. What is the purpose of kallsyms in kernel debugging?
31. How do you configure the kernel to support a custom CAN FD controller?
32. What steps are needed to implement a custom PWM controller driver?
33. How do you optimize the kernel for minimal interrupt latency in a multi-core system?
34. What is the role of devm_ functions in kernel memory management?
35. How do you configure the kernel for a specific FPGA-based peripheral?
36. What is the process to implement a custom power management IC driver?
37. How do you use kgdb for remote kernel debugging over Ethernet?
38. What steps are required to support a custom wireless chipset with proprietary firmware?
39. How do you implement a kernel module to handle a custom UART controller?
40. What is the purpose of slab versus slub allocators in the kernel?
41. How do you configure the kernel for a specific display driver with DRM/KMS?
42. What is the process to implement a custom voltage regulator driver?

43. How do you optimize the kernel for minimal memory fragmentation?
44. What steps are needed to support a custom secure element driver?
45. How do you configure the kernel to use a specific hardware video decoder?
46. What is the role of irqchip in advanced interrupt handling?
47. How do you implement a kernel module to manage a custom battery management system?
48. What is the process to configure the kernel for a specific touch controller?
49. How do you use systemtap to trace kernel performance?
50. What steps are required to support a custom I2C controller driver?
51. How do you configure the kernel to use a specific memory controller?
52. What is the purpose of kbuild in kernel module development?
53. How do you implement a kernel module to handle a custom high-resolution timer?
54. What is the process to optimize the kernel for minimal cache misses?
55. How do you configure the kernel for a specific Ethernet switch driver?
56. What steps are needed to support a custom Bluetooth controller with BLE?
57. How do you implement a kernel module to handle a custom GPIO expander?
58. What is the role of CONFIG_NO_HZ_FULL in real-time systems?
59. How do you configure the kernel for a specific LoRa controller?
60. What is the process to implement a custom watchdog timer driver?
61. How do you optimize the kernel for deterministic network performance?
62. What steps are required to support a custom MIPI DSI display driver?
63. How do you implement a kernel module to manage a custom power domain?
64. What is the purpose of trace-cmd in kernel tracing?
65. How do you configure the kernel for a specific hardware crypto accelerator?
66. What is the process to implement a custom SPI flash driver?
67. How do you optimize the kernel for NUMA-aware systems?
68. What steps are needed to support a custom I2S audio driver?
69. How do you implement a kernel module to handle a custom interrupt-driven sensor?
70. What is the role of kobject in kernel driver development?
71. How do you configure the kernel for a specific CANopen protocol stack?
72. What is the process to implement a custom RTC driver with PPS support?
73. How do you optimize the kernel for minimal context switch overhead?
74. What steps are required to support a custom PCIe endpoint driver?
75. How do you implement a kernel module to manage a custom clock gating mechanism?
76. What is the purpose of CONFIG_KASAN in kernel debugging?
77. How do you configure the kernel for a specific high-speed ADC driver?
78. What is the process to implement a custom memory-mapped peripheral driver?
79. How do you optimize the kernel for minimal TLB misses?
80. What steps are needed to support a custom secure boot chain with TPM?

Device Drivers and Low-Level Programming (80 Questions)

81. How do you write a character device driver to expose a hardware FIFO?
82. What is the process to implement a platform driver for an I2C sensor?
83. How do you write a block device driver for a custom NAND flash?
84. What steps are needed to implement a network driver for an Ethernet controller?
85. How do you write a USB gadget driver to emulate a mass storage device?
86. What is the process to implement a SPI driver for a multi-chip-select peripheral?
87. How do you write an input device driver for a custom touch panel?

88. What steps are required to implement a PWM driver for a motor controller?
89. How do you write a GPIO driver for a custom expander chip?
90. What is the process to implement a DMA driver for a hardware accelerator?
91. How do you write a watchdog driver for a custom hardware timer?
92. What steps are needed to implement a thermal driver for a temperature sensor?
93. How do you write a regulator driver for a custom PMIC?
94. What is the process to implement a clock driver for a PLL?
95. How do you write a pinctrl driver for a custom pin multiplexer?
96. What steps are required to implement a UART driver with flow control?
97. How do you write a CAN driver for a custom bus controller?
98. What is the process to implement an RTC driver for a high-precision clock?
99. How do you write a crypto driver for a hardware cryptographic engine?
100. What steps are needed to implement a display driver for a MIPI DSI panel?
101. How do you write a camera driver for a MIPI CSI-2 sensor?
102. What is the process to implement an audio driver for an I2S codec?
103. How do you write a USB driver for a custom USB 3.0 peripheral?
104. What steps are required to implement a PCIe driver for an endpoint device?
105. How do you write a memory driver for a custom SRAM controller?
106. What is the process to implement an interrupt controller driver?
107. How do you write a driver for a custom FPGA peripheral?
108. What steps are needed to implement a timer driver for a hardware timer?
109. How do you write a driver for a custom SPI NOR flash?
110. What is the process to implement a driver for an I2C EEPROM?
111. How do you write a driver for a custom high-speed ADC?
112. What steps are required to implement a driver for a high-precision DAC?
113. How do you write a driver for a custom multi-touch controller?
114. What is the process to implement a driver for a Wi-Fi module?
115. How do you write a driver for a custom GPS module with RTK?
116. What steps are needed to implement a driver for a LoRa module?
117. How do you write a driver for a custom Bluetooth module?
118. What is the process to implement a driver for an Ethernet switch?
119. How do you write a driver for a custom power monitor?
120. What steps are required to implement a driver for an accelerometer?
121. How do you write a driver for a custom gyroscope sensor?
122. What is the process to implement a driver for a magnetometer?
123. How do you write a driver for a custom proximity sensor?
124. What steps are needed to implement a driver for an ambient light sensor?
125. How do you write a driver for a custom pressure sensor?
126. What is the process to implement a driver for a humidity sensor?
127. How do you write a driver for a custom gas sensor?
128. What steps are required to implement a driver for a vibration sensor?
129. How do you write a driver for a custom thermal camera?
130. What is the process to implement a driver for an infrared sensor?
131. How do you write a driver for a custom CAN FD controller?
132. What steps are needed to implement a driver for a high-speed SPI controller?
133. How do you write a driver for a custom OLED display controller?
134. What is the process to implement a driver for a high-precision PWM?
135. How do you write a driver for a custom multi-channel ADC?
136. What steps are required to implement a driver for an I2C power monitor?

137. How do you write a driver for a custom high-resolution camera?
138. What is the process to implement a driver for a high-sensitivity gas sensor?
139. How do you write a driver for a custom high-speed UART?
140. What steps are needed to implement a driver for a LIDAR sensor?
141. How do you write a driver for a custom multi-port SPI controller?
142. What is the process to implement a driver for a high-precision IMU?
143. How do you write a driver for a custom I2C temperature sensor?
144. What steps are required to implement a driver for a high-speed DAC?
145. How do you write a driver for a custom ambient light sensor?
146. What is the process to implement a driver for a USB gadget with composite functions?
147. How do you write a driver for a custom Ethernet controller with PTP?
148. What steps are needed to implement a driver for a barometer?
149. How do you write a driver for a custom SPI-based touch controller?
150. What is the process to implement a driver for a high-speed ADC with oversampling?
151. How do you write a driver for a custom flow sensor?
152. What steps are required to implement a driver for an I2C EEPROM with secure write?
153. How do you write a driver for a custom high-speed UART with FIFO?
154. What is the process to implement a driver for a strain gauge?
155. How do you write a driver for a custom CAN bus controller?
156. What steps are needed to implement a driver for a high-resolution OLED?
157. How do you write a driver for a custom radar sensor?
158. What is the process to implement a driver for a SPI-based motor controller?
159. How do you write a driver for a custom high-precision timer?
160. What steps are required to implement a driver for a multi-port GPIO expander?

Networking and Communication (60 Questions)

161. How do you configure a device as a 5G NR gateway with QoS policies?
162. What is the process to implement a CoAP server with DTLS security?
163. How do you set up a device to use 6LoWPAN with RPL routing?
164. What steps are required to configure a Zigbee coordinator with custom firmware?
165. How do you implement a custom MQTT broker with TLS support?
166. What is the process to configure a Thread border router with NAT64?
167. How do you set up a LoRaWAN gateway with multi-channel support?
168. What steps are needed to implement a custom TCP/IP stack in a kernel module?
169. How do you configure DPDK for high-performance packet processing?
170. What is the process to set up a BACnet/IP server with custom objects?
171. How do you implement a UDP-based protocol for sensor streaming?
172. What steps are required to configure MPLS with traffic engineering?
173. How do you set up an OPC UA server with custom data models?
174. What is the process to implement a WebSocket server with binary streaming?
175. How do you configure SR-IOV for network virtualization?
176. What steps are needed to set up a DNP3 master for energy systems?
177. How do you implement a REST API server with OAuth2 authentication?
178. What is the process to configure CANopen with custom SDO handling?
179. How do you set up a Modbus TCP server with custom registers?
180. What steps are required to implement a gRPC service for device control?
181. How do you configure LwM2M for IoT device management?

182. What is the process to set up a KNX gateway with ETS integration?
183. How do you implement a ZeroMQ-based pub-sub system with encryption?
184. What steps are needed to configure TSN with time-aware scheduling?
185. How do you set up a Profinet controller for industrial automation?
186. What is the process to implement an HTTP/3 server for low-latency?
187. How do you configure GRE tunneling with IPsec encryption?
188. What steps are required to set up a BACnet MSTP router?
189. How do you implement a DDS publisher with QoS policies?
190. What is the process to configure VxLAN with EVPN for virtualization?
191. How do you set up a Z-Wave controller with S2 security?
192. What steps are needed to implement an RTSP server with H.265 streaming?
193. How do you configure Geneve tunneling with OVS integration?
194. What is the process to set up a LoRa gateway with custom forwarding?
195. How do you implement an SNMPv3 agent with USM and VACM?
196. What steps are required to configure IGMP snooping for multicast?
197. How do you set up an MQTT-SN gateway with QoS support?
198. What is the process to implement a SIP server with WebRTC integration?
199. How do you configure OpenFlow with custom flow rules?
200. What steps are needed to set up a CoAP proxy with caching?
201. How do you implement an NB-IoT client for low-power communication?
202. What is the process to configure mDNS with custom service advertisements?
203. How do you set up a Profibus master with custom DP-V1 support?
204. What steps are required to implement a gRPC server with streaming RPCs?
205. How do you configure LLDP for network topology discovery?
206. What is the process to set up a Thread border router with IPv6?
207. How do you implement a WebSocket-based control protocol with JWT?
208. What steps are needed to configure P4 for programmable packet processing?
209. How do you set up a Zigbee gateway with custom cluster support?
210. What is the process to implement an MQTT client with dynamic subscriptions?
211. How do you configure a device for high-availability networking with VRRP?
212. What steps are required to implement a custom FTP server with authentication?
213. How do you set up a device for 802.1X authentication?
214. What is the process to configure a device for multicast routing?
215. How do you implement a custom HTTP/2 server with push promises?
216. What steps are needed to set up a BACnet/IP client with custom objects?
217. How do you configure a device for network load balancing?
218. What is the process to implement a custom CoAP client with observe?
219. How do you set up a device for high-speed packet filtering with eBPF?
220. What steps are required to configure a device for IPv6 tunneling?

Real-Time Systems and Performance (60 Questions)

221. How do you configure a PREEMPT_RT kernel for sub-microsecond latency?
222. What is the process to implement a real-time control loop with SCHED_DEADLINE?
223. How do you optimize a system for minimal jitter in a 1kHz control loop?
224. What steps are needed to configure isolcpus for real-time tasks?
225. How do you implement a real-time data acquisition system with DMA?
226. What is the purpose of cyclictest, and how do you use it for latency testing?

227. How do you configure cpuset for task and interrupt isolation?
228. What steps are required to implement a real-time video pipeline with GStreamer?
229. How do you optimize a real-time system for minimal power consumption?
230. What is the process to configure SCHED_FIFO with priority inheritance?
231. How do you implement a real-time scheduler for a multi-threaded application?
232. What steps are needed to measure interrupt latency for a custom SPI device?
233. How do you configure PREEMPT_FULL for deterministic performance?
234. What is the process to implement a real-time logger for sensor data?
235. How do you optimize a real-time system for deterministic network communication?
236. What steps are required to configure irqaffinity for real-time interrupts?
237. How do you implement a real-time control system for a motor driver?
238. What is the process to measure and optimize context switch latency?
239. How do you configure SCHED_RR for a periodic control task?
240. What steps are needed to implement a real-time fault detection system?
241. How do you optimize a real-time system for minimal cache thrashing?
242. What is the process to configure cpufreq for real-time performance?
243. How do you implement a real-time watchdog for critical tasks?
244. What steps are required to measure DMA transfer latency in real-time?
245. How do you configure numactl for NUMA-aware real-time tasks?
246. What is the process to implement synchronized multi-sensor data collection?
247. How do you optimize a real-time system for minimal memory latency?
248. What steps are needed to configure taskset for core affinity?
249. How do you implement a real-time system for high-frequency PWM control?
250. What is the process to measure system call latency in a real-time application?
251. How do you configure rtirq for prioritizing real-time interrupts?
252. What steps are required to implement a real-time CAN bus data processor?
253. How do you optimize a real-time system for minimal TLB misses?
254. What is the process to configure cgroup for real-time resource limits?
255. How do you implement a real-time audio-video streaming system?
256. What steps are needed to measure interrupt coalescing latency?
257. How do you configure SCHED_DEADLINE for a multi-core control task?
258. What is the process to implement a real-time GNSS data processor?
259. How do you optimize a real-time system for minimal I/O latency?
260. What steps are required to configure rtkit for dynamic scheduling?
261. How do you implement a real-time system for high-speed SPI data acquisition?
262. What is the process to measure memory bandwidth in a real-time application?
263. How do you configure nohz_full for tickless real-time operation?
264. What steps are needed to implement synchronized multi-camera capture?
265. How do you optimize a real-time system for minimal idle power?
266. What is the process to configure cpuidle for real-time power management?
267. How do you implement a real-time system for high-frequency I2C polling?
268. What steps are required to measure cache coherence overhead?
269. How do you configure PREEMPT_RT with custom latency tuning?
270. What is the process to implement a real-time motor control with PID?
271. How do you optimize a real-time system for minimal memory fragmentation?
272. What steps are needed to configure a real-time system for high-speed UART?
273. How do you implement a real-time system for synchronized sensor fusion?
274. What is the process to measure and optimize scheduler latency?
275. How do you configure a device for real-time Ethernet with TSN?

276. What steps are required to implement a real-time data logger with buffering?
277. How do you optimize a real-time system for minimal interrupt overhead?
278. What is the process to configure a device for low-latency video processing?
279. How do you implement a real-time system for high-precision timing?
280. What steps are needed to measure and optimize real-time task preemption?

Security and Hardening (60 Questions)

281. How do you configure SELinux with a custom policy for an IoT application?
282. What is the process to implement a secure boot chain with TPM 2.0?
283. How do you configure AppArmor with a custom profile for a network service?
284. What steps are required to implement a kernel module for runtime MAC policies?
285. How do you configure dm-verity for root filesystem integrity?
286. What is the process to implement a secure OTA update with swupdate?
287. How do you configure seccomp with a custom syscall whitelist?
288. What steps are needed to implement a TEE driver for secure operations?
289. How do you configure IMA/EVM with a custom keyring?
290. What is the process to implement a secure key storage with an HSM?
291. How do you configure nftables with a stateful firewall policy?
292. What steps are required to implement an encryption driver for storage?
293. How do you configure WireGuard with post-quantum cryptography?
294. What is the process to implement a tamper-proof logging system?
295. How do you configure Smack with a custom MAC policy?
296. What steps are needed to implement a kernel module for syscall monitoring?
297. How do you configure TOMOYO Linux with a custom security domain?
298. What is the process to implement a secure bootloader with signature verification?
299. How do you configure eCryptfs with a custom key management system?
300. What steps are required to implement process sandboxing with namespaces?
301. How do you configure grsecurity with custom PaX flags?
302. What is the process to implement remote attestation with TPM?
303. How do you configure auditd with custom intrusion detection rules?
304. What steps are needed to implement a kernel module for rootkit detection?
305. How do you configure LUKS with a custom key derivation function?
306. What is the process to implement a secure TLS communication channel?
307. How do you configure seccomp-bpf for application sandboxing?
308. What steps are required to implement network access control with eBPF?
309. How do you configure Yama LSM for enhanced ptrace security?
310. What is the process to implement a secure firmware update system?
311. How do you configure KASLR with a custom randomization algorithm?
312. What steps are needed to implement a kernel module for memory encryption?
313. How do you configure dm-crypt with a custom cipher suite?
314. What is the process to implement a secure boot with encrypted kernel images?
315. How do you configure AppArmor with a custom network policy?
316. What steps are required to implement a kernel module for privilege monitoring?
317. How do you configure SELinux with a custom MLS policy?
318. What is the process to implement a secure key exchange with hardware RNG?
319. How do you configure nftables with a custom NAT policy for IoT?
320. What steps are needed to implement runtime integrity checking?

321. How do you configure IMA with a custom appraisal policy?
322. What is the process to implement secure remote management with SSH and TOTP?
323. How do you configure TOMOYO with a custom path-based policy?
324. What steps are required to implement secure IPC in a kernel module?
325. How do you configure seccomp for a web server sandbox?
326. What is the process to implement secure data storage with hardware encryption?
327. How do you configure grsecurity with a custom RBAC policy?
328. What steps are needed to implement a kernel module for exploit detection?
329. How do you configure LUKS with a custom keyslot management system?
330. What is the process to implement a secure boot chain with U-Boot?
331. How do you configure a device to use AppArmor with a custom filesystem policy?
332. What steps are required to implement a secure logging system with encryption?
333. How do you configure SELinux for a containerized application?
334. What is the process to implement a secure boot with verified initramfs?
335. How do you configure nftables for advanced packet filtering?
336. What steps are needed to implement a kernel module for secure device access?
337. How do you configure IMA/EVM for runtime file verification?
338. What is the process to implement a secure communication protocol with mTLS?
339. How do you configure seccomp for a specific kernel module?
340. What steps are required to implement a secure OTA update with anti-rollback?

Hardware Interfacing and Peripherals (60 Questions)

341. How do you interface a high-speed ADC with a custom DMA driver?
342. What is the process to configure a MIPI CSI-2 camera with 4K support?
343. How do you implement a driver for a high-resolution e-ink display?
344. What steps are required to interface a GNSS module with RTK support?
345. How do you configure a PCIe endpoint driver with DMA transfers?
346. What is the process to implement a high-speed UART driver?
347. How do you interface a high-frequency accelerometer with interrupts?
348. What steps are needed to configure a SPI NOR flash with quad-SPI?
349. How do you implement a driver for a multi-channel audio codec?
350. What is the process to interface a thermal camera with real-time processing?
351. How do you configure an I2C multiplexer with dynamic routing?
352. What steps are required to implement a motor controller driver with torque control?
353. How do you interface a high-speed CAN FD bus with error correction?
354. What is the process to configure a PWM fan controller with PID?
355. How do you implement a driver for a multi-touch controller with pressure?
356. What steps are needed to interface a magnetometer with calibration algorithms?
357. How do you configure a USB 3.1 device driver with SuperSpeed support?
358. What is the process to implement a driver for an FPGA peripheral with AXI?
359. How do you interface a high-speed Ethernet switch with VLAN support?
360. What steps are required to configure a MIPI DSI display with adaptive refresh?
361. How do you implement a driver for a high-precision RTC with PPS?
362. What is the process to interface a LoRa module with custom modulation?
363. How do you configure an I2S audio driver with multi-stream support?
364. What steps are needed to implement a multi-port GPIO expander driver?
365. How do you interface a proximity sensor with adaptive ranging?

366. What is the process to configure a CANopen driver with PDO mappings?
367. How do you implement a driver for a high-speed SPI controller?
368. What steps are required to interface a pressure sensor with compensation?
369. How do you configure a USB-C PD controller with fast role swap?
370. What is the process to implement a high-speed ADC driver with DMA?
371. How do you interface a gyroscope with real-time sensor fusion?
372. What steps are needed to configure a PCIe switch driver with hot-plug?
373. How do you implement a driver for a high-speed UART with FIFO?
374. What is the process to interface a LIDAR sensor with point cloud processing?
375. How do you configure an I2C-based power monitor with event triggers?
376. What steps are required to implement a high-resolution camera driver?
377. How do you interface a gas sensor with calibration routines?
378. What is the process to configure a SPI-based display controller?
379. How do you implement a driver for a high-precision PWM controller?
380. What steps are needed to interface a vibration sensor with FFT analysis?
381. How do you configure a CAN bus controller with extended frames?
382. What is the process to implement a SPI flash driver with ECC?
383. How do you interface an IMU with 9-DOF fusion?
384. What steps are required to configure an I2C temperature sensor with multi-zone support?
385. How do you implement a driver for a high-speed DAC with waveform generation?
386. What is the process to interface an ambient light sensor with spectral compensation?
387. How do you configure a USB gadget driver with multi-function support?
388. What steps are needed to implement an Ethernet controller driver with PTP?
389. How do you interface a barometer with altitude tracking algorithms?
390. What is the process to configure a SPI-based touch controller with haptics?
391. How do you implement a driver for a high-speed ADC with oversampling?
392. What steps are required to interface a flow sensor with flow rate calculation?
393. How do you configure an I2C-based EEPROM with secure write protection?
394. What is the process to implement a high-speed UART driver with DMA?
395. How do you interface a strain gauge with real-time signal processing?
396. What steps are needed to configure a CAN FD driver with custom error handling?
397. How do you implement a driver for a high-resolution OLED with gamma correction?
398. What is the process to interface a radar sensor with Doppler processing?
399. How do you configure a SPI-based motor controller with closed-loop control?
400. What steps are required to implement a high-precision timer driver?

Notes

- These questions are designed for advanced candidates with deep expertise in embedded Linux, including kernel development, device drivers, networking, real-time systems, security, and hardware interfacing.
- Candidates should be proficient with tools like Buildroot, Yocto, gcc, gdb, perf, ftrace, and hardware debugging tools (e.g., JTAG, oscilloscopes).
- Practical preparation requires hands-on experience with embedded platforms, cross-compilation, and advanced debugging techniques.
- Many questions involve custom hardware, so candidates should have access to or simulate hardware using QEMU or similar tools.

500 Expert-Level Embedded Linux Interview Questions

Kernel Development and Optimization (100 Questions)

1. How do you implement a custom kernel scheduler for a specific real-time workload?
2. What steps are required to optimize the Linux kernel for sub-100μs interrupt latency?
3. How do you configure the kernel to support a custom NUMA architecture?
4. What is the process to implement a kernel module for runtime memory encryption?
5. How do you optimize the kernel for minimal boot time (<50ms) on an ARM SoC?
6. What is the role of CONFIG_KASAN in detecting kernel memory issues?
7. How do you implement a custom power management framework for a multi-core SoC?
8. What steps are needed to configure the kernel for a custom PCIe root complex with SR-IOV?
9. How do you use ftrace to profile kernel performance under a high-frequency interrupt load?
10. What is the process to implement a custom clocksource driver for a high-precision timer?
11. How do you configure the kernel to support a custom GPU with DMA-BUF integration?
12. What steps are required to implement a kernel module for a custom cryptographic accelerator?
13. How do you optimize the kernel for minimal TLB misses in a multi-threaded application?
14. What is the role of kprobes in dynamic kernel instrumentation?
15. How do you configure the kernel for a custom Ethernet controller with PTP support?
16. What steps are needed to implement a kernel module for a custom thermal zone with hysteresis?
17. How do you use perf to analyze cache coherence issues in a multi-core system?
18. What is the process to configure the kernel for a custom MIPI CSI-2 camera with 8K support?
19. How do you implement a kernel module for a custom interrupt controller with nested IRQs?
20. What steps are required to optimize the kernel for minimal context switch overhead?
21. How do you configure the kernel to support a custom FPGA with AXI interconnects?
22. What is the role of kcov in kernel code coverage analysis?
23. How do you implement a kernel module for a custom high-speed ADC with DMA?
24. What steps are needed to configure the kernel for a custom LoRa controller with ECC?
25. How do you optimize the kernel for minimal memory fragmentation in a long-running system?
26. What is the process to implement a kernel module for a custom power domain controller?
27. How do you configure the kernel to support a custom USB-C controller with PD 3.0?
28. What steps are required to implement a kernel module for a custom SPI flash with QSPI?
29. How do you use systemtap to trace kernel performance in a real-time system?
30. What is the process to configure the kernel for a custom CAN FD controller with ISO 16845?
31. How do you implement a kernel module for a custom high-resolution PWM controller?
32. What steps are needed to optimize the kernel for NUMA-aware memory allocation?
33. How do you configure the kernel to support a custom MIPI DSI display with HDR?
34. What is the process to implement a kernel module for a custom RTC with PPS and GPS sync?
35. How do you optimize the kernel for minimal cache thrashing in a multi-core system?
36. What steps are required to configure the kernel for a custom wireless chipset with 802.11ax?
37. How do you implement a kernel module for a custom high-speed UART with FIFO?
38. What is the role of CONFIG_NO_HZ_FULL in tickless real-time systems?
39. How do you configure the kernel to support a custom audio codec with multi-stream?
40. What steps are needed to implement a kernel module for a custom voltage regulator?
41. How do you optimize the kernel for minimal power consumption in a battery-powered IoT device?
42. What is the process to configure the kernel for a custom I2C controller with SMBus support?

43. How do you implement a kernel module for a custom high-precision timer with 1ns resolution?
44. What steps are required to configure the kernel for a custom Ethernet switch with TSN?
45. How do you optimize the kernel for deterministic network performance in a 5G gateway?
46. What is the process to implement a kernel module for a custom memory controller with ECC?
47. How do you configure the kernel to support a custom PCIe endpoint with MSI-X?
48. What steps are needed to implement a kernel module for a custom thermal camera driver?
49. How do you use kgdb for remote debugging of a kernel crash over Ethernet?
50. What is the process to configure the kernel for a custom GNSS receiver with RTK?
51. How do you implement a kernel module for a custom clock gating mechanism?
52. What steps are required to optimize the kernel for minimal interrupt coalescing latency?
53. How do you configure the kernel to support a custom high-speed SPI controller?
54. What is the process to implement a kernel module for a custom power monitor with alerts?
55. How do you optimize the kernel for minimal memory latency in a real-time system?
56. What steps are needed to configure the kernel for a custom I2S audio driver with 192kHz?
57. How do you implement a kernel module for a custom multi-port GPIO expander?
58. What is the role of devm_ functions in kernel resource management?
59. How do you configure the kernel to support a custom display driver with V4L2?
60. What steps are required to implement a kernel module for a custom high-speed DAC?
61. How do you optimize the kernel for minimal scheduler latency in a multi-core system?
62. What is the process to configure the kernel for a custom Bluetooth controller with BLE 5.2?
63. How do you implement a kernel module for a custom high-frequency accelerometer?
64. What steps are needed to configure the kernel for a custom CANopen stack with PDO?
65. How do you optimize the kernel for minimal I/O latency in a storage driver?
66. What is the process to implement a kernel module for a custom LIDAR sensor driver?
67. How do you configure the kernel to support a custom USB 3.1 gadget with SuperSpeed?
68. What steps are required to implement a kernel module for a custom radar sensor?
69. How do you optimize the kernel for minimal cache misses in a real-time application?
70. What is the process to configure the kernel for a custom high-speed ADC with oversampling?
71. How do you implement a kernel module for a custom pressure sensor with compensation?
72. What steps are needed to configure the kernel for a custom Ethernet controller with 10GbE?
73. How do you optimize the kernel for minimal memory bandwidth contention?
74. What is the process to implement a kernel module for a custom I2C-based power monitor?
75. How do you configure the kernel to support a custom multi-channel audio driver?
76. What steps are required to implement a kernel module for a custom high-resolution IMU?
77. How do you optimize the kernel for minimal interrupt overhead in a multi-core system?
78. What is the process to configure the kernel for a custom SPI-based touch controller?
79. How do you implement a kernel module for a custom high-precision temperature sensor?
80. What steps are needed to configure the kernel for a custom PCIe switch with hot-plug?
81. How do you optimize the kernel for minimal power consumption in a multi-core SoC?
82. What is the process to implement a kernel module for a custom vibration sensor with FFT?
83. How do you configure the kernel to support a custom high-speed UART with DMA?
84. What steps are required to implement a kernel module for a custom gas sensor with calibration?
85. How do you optimize the kernel for minimal context switch latency in a real-time system?
86. What is the process to configure the kernel for a custom LoRaWAN controller?
87. How do you implement a kernel module for a custom high-speed SPI flash driver?
88. What steps are needed to configure the kernel for a custom MIPI CSI-2 camera with HDR?
89. How do you optimize the kernel for minimal memory allocation latency?
90. What is the process to implement a kernel module for a custom ambient light sensor?
91. How do you configure the kernel to support a custom CAN bus controller with error handling?

92. What steps are required to implement a kernel module for a custom high-resolution OLED?
93. How do you optimize the kernel for minimal TLB thrashing in a NUMA system?
94. What is the process to configure the kernel for a custom I2C-based EEPROM with secure write?
95. How do you implement a kernel module for a custom high-speed DAC with waveform generation?
96. What steps are needed to configure the kernel for a custom Ethernet controller with QoS?
97. How do you optimize the kernel for minimal interrupt latency in a 5G NR gateway?
98. What is the process to implement a kernel module for a custom multi-port SPI controller?
99. How do you configure the kernel to support a custom high-precision GNSS receiver?
100. What steps are required to implement a kernel module for a custom flow sensor with rate calculation?

Device Drivers and Low-Level Programming (100 Questions)

101. How do you implement a character device driver for a custom hardware FIFO with atomic operations?
102. What is the process to write a platform driver for a multi-channel I2C sensor?
103. How do you implement a block device driver for a custom NAND flash with ECC?
104. What steps are required to write a network driver for a custom 10GbE controller?
105. How do you implement a USB gadget driver for a composite device with CDC and HID?
106. What is the process to write a SPI driver for a multi-chip-select peripheral with DMA?
107. How do you implement an input device driver for a custom capacitive touch panel?
108. What steps are needed to write a PWM driver for a high-frequency motor controller?
109. How do you implement a GPIO driver for a custom 32-bit expander with interrupts?
110. What is the process to write a DMA driver for a custom hardware accelerator with scatter-gather?
111. How do you implement a watchdog driver for a custom hardware timer with reset?
112. What steps are required to write a thermal driver for a multi-zone temperature sensor?
113. How do you implement a regulator driver for a custom PMIC with dynamic voltage scaling?
114. What is the process to write a clock driver for a custom PLL with fractional divider?
115. How do you implement a pinctrl driver for a custom pin multiplexer with dynamic mapping?
116. What steps are needed to write a UART driver with hardware flow control and FIFO?
117. How do you implement a CAN driver for a custom bus controller with error injection?
118. What is the process to write an RTC driver for a high-precision clock with alarm?
119. How do you implement a crypto driver for a custom AES hardware engine?
120. What steps are required to write a display driver for a MIPI DSI panel with adaptive refresh?
121. How do you implement a camera driver for a MIPI CSI-2 sensor with 4K and HDR?
122. What is the process to write an audio driver for an I2S codec with multi-stream support?
123. How do you implement a USB driver for a custom USB 3.1 peripheral with Isochronous transfers?
124. What steps are needed to write a PCIe driver for an endpoint device with MSI-X?
125. How do you implement a memory driver for a custom SRAM controller with ECC?
126. What is the process to write an interrupt controller driver with priority support?
127. How do you implement a driver for a custom FPGA peripheral with AXI4-Lite?
128. What steps are required to write a timer driver for a custom hardware timer with capture?
129. How do you implement a driver for a custom SPI NOR flash with dual/quad modes?
130. What is the process to write a driver for an I2C EEPROM with secure write protection?
131. How do you implement a driver for a custom high-speed ADC with oversampling and DMA?
132. What steps are needed to write a driver for a high-precision DAC with waveform generation?
133. How do you implement a driver for a custom multi-touch controller with pressure sensitivity?
134. What is the process to write a driver for a Wi-Fi module with 802.11ax support?

135. How do you implement a driver for a custom GPS module with RTK and NMEA?
136. What steps are required to write a driver for a LoRa module with custom modulation?
137. How do you implement a driver for a custom Bluetooth module with BLE 5.2?
138. What is the process to write a driver for an Ethernet switch with VLAN and QoS?
139. How do you implement a driver for a custom power monitor with event triggers?
140. What steps are needed to write a driver for an accelerometer with high-frequency sampling?
141. How do you implement a driver for a custom gyroscope with 9-DOF fusion?
142. What is the process to write a driver for a magnetometer with automatic calibration?
143. How do you implement a driver for a custom proximity sensor with adaptive ranging?
144. What steps are required to write a driver for an ambient light sensor with spectral correction?
145. How do you implement a driver for a custom pressure sensor with temperature compensation?
146. What is the process to write a driver for a humidity sensor with dew point calculation?
147. How do you implement a driver for a custom gas sensor with calibration algorithms?
148. What steps are needed to write a driver for a vibration sensor with FFT analysis?
149. How do you implement a driver for a custom thermal camera with real-time processing?
150. What is the process to write a driver for an infrared sensor with gesture detection?
151. How do you implement a driver for a custom CAN FD controller with error handling?
152. What steps are required to write a driver for a high-speed SPI controller with DMA?
153. How do you implement a driver for a custom OLED display with gamma correction?
154. What is the process to write a driver for a high-precision PWM controller with dead-time?
155. How do you implement a driver for a custom multi-channel ADC with noise filtering?
156. What steps are needed to write a driver for an I2C-based power monitor with alerts?
157. How do you implement a driver for a custom high-resolution camera with RAW output?
158. What is the process to write a driver for a high-sensitivity gas sensor with calibration?
159. How do you implement a driver for a custom high-speed UART with DMA and FIFO?
160. What steps are required to write a driver for a LIDAR sensor with point cloud processing?
161. How do you implement a driver for a custom multi-port SPI controller with priority queuing?
162. What is the process to write a driver for a high-precision IMU with sensor fusion?
163. How do you implement a driver for a custom I2C temperature sensor with multi-zone support?
164. What steps are needed to write a driver for a high-speed DAC with multi-channel support?
165. How do you implement a driver for a custom ambient light sensor with adaptive gain?
166. What is the process to write a driver for a USB gadget with multi-function composite?
167. How do you implement a driver for a custom Ethernet controller with IEEE 1588?
168. What steps are required to write a driver for a barometer with altitude tracking?
169. How do you implement a driver for a custom SPI-based touch controller with haptics?
170. What is the process to write a driver for a high-speed ADC with dynamic range control?
171. How do you implement a driver for a custom flow sensor with rate calculation?
172. What steps are needed to write a driver for an I2C-based EEPROM with secure erase?
173. How do you implement a driver for a custom high-speed UART with error correction?
174. What is the process to write a driver for a strain gauge with real-time processing?
175. How do you implement a driver for a custom CAN bus controller with extended frames?
176. What steps are required to write a driver for a high-resolution OLED with color calibration?
177. How do you implement a driver for a custom radar sensor with Doppler processing?
178. What is the process to write a driver for a SPI-based motor controller with closed-loop control?
179. How do you implement a driver for a custom high-precision timer with event capture?
180. What steps are needed to write a driver for a multi-port GPIO expander with interrupt chaining?
181. How do you implement a driver for a custom high-frequency accelerometer with DMA?
182. What is the process to write a driver for a magnetometer with geomagnetic compensation?
183. How do you implement a driver for a custom proximity sensor with multi-zone detection?

184. What steps are required to write a driver for an ambient light sensor with IR rejection?
185. How do you implement a driver for a custom pressure sensor with high-precision calibration?
186. What is the process to write a driver for a humidity sensor with temperature correction?
187. How do you implement a driver for a custom gas sensor with multi-gas detection?
188. What steps are needed to write a driver for a vibration sensor with real-time analysis?
189. How do you implement a driver for a custom thermal camera with HDR processing?
190. What is the process to write a driver for an infrared sensor with proximity detection?
191. How do you implement a driver for a custom CAN FD controller with dynamic bitrate?
192. What steps are required to write a driver for a high-speed SPI controller with ECC?
193. How do you implement a driver for a custom OLED display with adaptive brightness?
194. What is the process to write a driver for a high-precision PWM controller with phase shift?
195. How do you implement a driver for a custom multi-channel ADC with differential inputs?
196. What steps are needed to write a driver for an I2C-based power monitor with dynamic range?
197. How do you implement a driver for a custom high-resolution camera with YUV output?
198. What is the process to write a driver for a high-sensitivity gas sensor with drift compensation?
199. How do you implement a driver for a custom high-speed UART with multi-port support?
200. What steps are required to write a driver for a LIDAR sensor with real-time mapping?

Networking and Communication (80 Questions)

201. How do you configure a device as a 5G NR gateway with custom QoS and network slicing?
202. What is the process to implement a CoAP server with DTLS 1.3 and PSK authentication?
203. How do you set up a device for 6LoWPAN with custom RPL routing policies?
204. What steps are required to configure a Zigbee coordinator with custom ZCL profiles?
205. How do you implement an MQTT broker with TLS 1.3 and dynamic QoS?
206. What is the process to configure a Thread border router with IPv6 and NAT64?
207. How do you set up a LoRaWAN gateway with custom packet forwarding and encryption?
208. What steps are needed to implement a custom TCP/IP stack in a kernel module with zero-copy?
209. How do you configure DPDK for high-performance packet processing with multi-core support?
210. What is the process to set up a BACnet/IP server with custom COV notifications?
211. How do you implement a UDP-based protocol for real-time sensor streaming with FEC?
212. What steps are required to configure MPLS with custom traffic engineering policies?
213. How do you set up an OPC UA server with custom information models and security?
214. What is the process to implement a WebSocket server with binary streaming and compression?
215. How do you configure SR-IOV for high-performance network virtualization?
216. What steps are needed to set up a DNP3 master with custom polling strategies?
217. How do you implement a REST API server with OAuth2 and rate limiting?
218. What is the process to configure CANopen with custom SDO and PDO mappings?
219. How do you set up a Modbus TCP server with custom function codes?
220. What steps are required to implement a gRPC service with bidirectional streaming?
221. How do you configure LwM2M for IoT device management with custom objects?
222. What is the process to set up a KNX gateway with custom group addresses?
223. How do you implement a ZeroMQ-based pub-sub system with curve encryption?
224. What steps are needed to configure TSN with custom time-aware shaping?
225. How do you set up a Profinet controller with custom GSDML files?
226. What is the process to implement an HTTP/3 server with QPACK compression?
227. How do you configure GRE tunneling with IPsec and custom key management?
228. What steps are required to set up a BACnet MSTP router with custom routing?

229. How do you implement a DDS publisher with custom QoS and partitioning?
230. What is the process to configure VxLAN with EVPN and custom BGP policies?
231. How do you set up a Z-Wave controller with S2 security and custom commands?
232. What steps are needed to implement an RTSP server with H.265 and low-latency streaming?
233. How do you configure Geneve tunneling with OVS and custom metadata?
234. What is the process to set up a LoRa gateway with custom channel plans?
235. How do you implement an SNMPv3 agent with custom MIBs and USM?
236. What steps are required to configure IGMP snooping with custom multicast groups?
237. How do you set up an MQTT-SN gateway with custom topic mapping?
238. What is the process to implement a SIP server with WebRTC and ICE support?
239. How do you configure OpenFlow with custom flow tables and actions?
240. What steps are needed to set up a CoAP proxy with custom caching policies?
241. How do you implement an NB-IoT client with custom LWM2M objects?
242. What is the process to configure mDNS with custom service advertisements?
243. How do you set up a Profibus master with custom DP-V1 services?
244. What steps are required to implement a gRPC server with custom interceptors?
245. How do you configure LLDP with custom TLVs for topology discovery?
246. What is the process to set up a Thread border router with custom IPv6 routing?
247. How do you implement a WebSocket-based protocol with JWT and compression?
248. What steps are needed to configure P4 for custom programmable packet processing?
249. How do you set up a Zigbee gateway with custom binding tables?
250. What is the process to implement an MQTT client with custom will messages?
251. How do you configure a device for high-availability with VRRP and custom failover?
252. What steps are required to implement a custom FTP server with SFTP support?
253. How do you set up a device for 802.1X with custom EAP methods?
254. What is the process to configure multicast routing with PIM-SM?
255. How do you implement a custom HTTP/2 server with server push and prioritization?
256. What steps are needed to set up a BACnet/IP client with custom object mappings?
257. How do you configure a device for network load balancing with custom algorithms?
258. What is the process to implement a custom CoAP client with block-wise transfers?
259. How do you set up a device for high-speed packet filtering with eBPF and XDP?
260. What steps are required to configure IPv6 tunneling with custom encapsulation?
261. How do you implement a custom MQTT broker with dynamic ACLs?
262. What is the process to configure a device for 5G NR with custom DRB mappings?
263. How do you set up a Thread gateway with custom CoAP integration?
264. What steps are needed to implement a WebSocket server with custom subprotocols?
265. How do you configure a device for SRv6 with custom segment routing?
266. What is the process to set up a LoRaWAN server with custom authentication?
267. How do you implement a custom UDP protocol with congestion control?
268. What steps are required to configure a Zigbee coordinator with custom OTA updates?
269. How do you set up a device for CANopen with custom NMT state handling?
270. What is the process to implement a REST API with custom rate limiting and caching?
271. How do you configure a device for high-performance networking with DPDK and AF_XDP?
272. What steps are needed to set up a Profinet device with custom cyclic data?
273. How do you implement a custom CoAP server with resource discovery?
274. What is the process to configure a device for MPLS with custom label switching?
275. How do you set up a BACnet router with custom virtual networks?
276. What steps are required to implement a DDS subscriber with custom filters?
277. How do you configure a device for 802.11ax with custom MU-MIMO?

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278. What is the process to set up an MQTT gateway with custom bridge connections?
279. How do you implement a custom SIP client with SRTP encryption?
280. What steps are needed to configure a device for TSN with custom AVB streams?

Real-Time Systems and Performance (80 Questions)

281. How do you configure a PREEMPT_RT kernel for sub-microsecond latency in a multi-core system?
282. What is the process to implement a real-time control loop with SCHED_DEADLINE and cgroups?
283. How do you optimize a system for minimal jitter in a 10kHz control loop?
284. What steps are required to configure isolcpus and irqaffinity for real-time tasks?
285. How do you implement a real-time data acquisition system with DMA and zero-copy?
286. What is the purpose of cyclictest, and how do you use it for sub-microsecond latency testing?
287. How do you configure cpuset for task and interrupt isolation in a NUMA system?
288. What steps are needed to implement a real-time video pipeline with GStreamer and V4L2?
289. How do you optimize a real-time system for minimal power consumption with DVFS?
290. What is the process to configure SCHED_FIFO with mutex priority inheritance?
291. How do you implement a real-time scheduler for a multi-threaded control application?
292. What steps are required to measure interrupt latency for a custom PCIe device?
293. How do you configure PREEMPT_FULL with custom preempt points for determinism?
294. What is the process to implement a real-time logger for high-frequency sensor data?
295. How do you optimize a real-time system for deterministic network communication with TSN?
296. What steps are needed to configure rtirq for prioritizing real-time interrupts?
297. How do you implement a real-time control system for a multi-axis motor driver?
298. What is the process to measure and optimize context switch latency in a real-time system?
299. How do you configure SCHED_RR for a periodic control task with sub-ms precision?
300. What steps are required to implement a real-time fault detection system with failover?
301. How do you optimize a real-time system for minimal cache coherence overhead?
302. What is the process to configure cpufreq for real-time performance with custom governors?
303. How do you implement a real-time watchdog for mission-critical tasks?
304. What steps are needed to measure DMA transfer latency in a real-time application?
305. How do you configure numactl for NUMA-aware real-time task scheduling?
306. What is the process to implement synchronized multi-sensor data collection with PPS?
307. How do you optimize a real-time system for minimal memory access latency?
308. What steps are required to configure taskset for dynamic CPU affinity?
309. How do you implement a real-time system for high-frequency PWM control with 1µs precision?
310. What is the process to measure system call latency in a real-time application?
311. How do you configure nohz_full for tickless operation in a real-time system?
312. What steps are needed to implement a real-time CAN bus data processor with sub-ms latency?
313. How do you optimize a real-time system for minimal TLB misses in a multi-core setup?
314. What is the process to configure cgroup for real-time resource isolation?
315. How do you implement a real-time audio-video streaming system with <10ms latency?
316. What steps are required to measure interrupt coalescing latency in a real-time system?
317. How do you configure SCHED_DEADLINE for a multi-core control task with deadlines?
318. What is the process to implement a real-time GNSS data processor with RTK?
319. How do you optimize a real-time system for minimal I/O latency with DMA?
320. What steps are needed to configure rtkit for dynamic real-time scheduling?
321. How do you implement a real-time system for high-speed SPI data acquisition?
322. What is the process to measure memory bandwidth in a real-time application?

323. How do you configure a real-time system for low-latency video processing with V4L2?
324. What steps are required to implement synchronized multi-camera capture with sub-ms sync?
325. How do you optimize a real-time system for minimal idle power consumption?
326. What is the process to configure cpuidle for real-time power management?
327. How do you implement a real-time system for high-frequency I2C polling with 1kHz rate?
328. What steps are needed to measure cache coherence overhead in a real-time system?
329. How do you configure PREEMPT_RT with custom latency tuning for a 5G gateway?
330. What is the process to implement a real-time motor control system with PID and feedforward?
331. How do you optimize a real-time system for minimal memory fragmentation in long-running tasks?
332. What steps are required to configure a real-time system for high-speed UART with DMA?
333. How do you implement a real-time system for synchronized sensor fusion with Kalman filtering?
334. What is the process to measure and optimize scheduler latency in a multi-core system?
335. How do you configure a device for real-time Ethernet with TSN and AVB?
336. What steps are needed to implement a real-time data logger with circular buffering?
337. How do you optimize a real-time system for minimal interrupt overhead with MSI?
338. What is the process to configure a device for low-latency video encoding with hardware acceleration?
339. How do you implement a real-time system for high-precision timing with 1ns resolution?
340. What steps are required to measure and optimize real-time task preemption latency?
341. How do you configure a real-time system for high-speed SPI with zero-copy transfers?
342. What is the process to implement a real-time control loop with adaptive gain?
343. How do you optimize a real-time system for minimal memory bandwidth contention?
344. What steps are needed to configure a real-time system for synchronized ADC sampling?
345. How do you implement a real-time system for high-frequency PWM with phase alignment?
346. What is the process to measure and optimize interrupt latency for a custom CAN controller?
347. How do you configure a real-time system for low-latency audio with ALSA?
348. What steps are required to implement a real-time fault-tolerant control system?
349. How do you optimize a real-time system for minimal cache thrashing with prefetching?
350. What is the process to configure a real-time system for high-speed I2C with multi-master?
351. How do you implement a real-time system for high-precision GNSS with SBAS?
352. What steps are needed to measure and optimize memory access latency in a real-time system?
353. How do you configure a real-time system for deterministic network communication with DPDK?
354. What is the process to implement a real-time video pipeline with hardware decoding?
355. How do you optimize a real-time system for minimal power consumption with C-states?
356. What steps are required to configure a real-time system for synchronized sensor polling?
357. How do you implement a real-time system for high-speed UART with error correction?
358. What is the process to measure and optimize interrupt coalescing in a real-time system?
359. How do you configure a real-time system for low-latency video streaming with RTSP?
360. What steps are needed to implement a real-time control system with redundant sensors?

Security and Hardening (80 Questions)

361. How do you configure SELinux with a custom MLS policy for an IoT device?
362. What is the process to implement a secure boot chain with TPM 2.0 and measured boot?
363. How do you configure AppArmor with a custom profile for a network daemon?
364. What steps are required to implement a kernel module for runtime syscall filtering?
365. How do you configure dm-verity with a custom hash tree for filesystem integrity?
366. What is the process to implement a secure OTA update with swupdate and anti-rollback?

367. How do you configure seccomp with a custom syscall whitelist for a container?
368. What steps are needed to implement a TEE driver for secure key storage?
369. How do you configure IMA/EVM with a custom keyring and appraisal policy?
370. What is the process to implement a secure key management system with HSM?
371. How do you configure nftables with a custom stateful firewall for IoT traffic?
372. What steps are required to implement an encryption driver for secure storage?
373. How do you configure WireGuard with post-quantum cryptography and custom keys?
374. What is the process to implement a tamper-proof logging system with blockchain?
375. How do you configure Smack with a custom MAC policy for embedded applications?
376. What steps are needed to implement a kernel module for runtime privilege monitoring?
377. How do you configure TOMOYO Linux with a custom path-based security policy?
378. What is the process to implement a secure bootloader with signature verification and rollback?
379. How do you configure eCryptfs with a custom key derivation function?
380. What steps are required to implement process sandboxing with namespaces and cgroups?
381. How do you configure grsecurity with custom PaX flags and RBAC?
382. What is the process to implement remote attestation with TPM and secure boot?
383. How do you configure auditd with custom rules for intrusion detection?
384. What steps are needed to implement a kernel module for rootkit detection?
385. How do you configure LUKS with a custom keyslot and anti-brute-force measures?
386. What is the process to implement a secure TLS communication channel with mTLS?
387. How do you configure seccomp-bpf for a high-security web server?
388. What steps are required to implement network access control with eBPF and XDP?
389. How do you configure Yama LSM for enhanced ptrace and process security?
390. What is the process to implement a secure firmware update system with encryption?
391. How do you configure KASLR with a custom randomization algorithm for kernel security?
392. What steps are needed to implement a kernel module for memory encryption with SGX?
393. How do you configure dm-crypt with a custom cipher suite and key rotation?
394. What is the process to implement a secure boot with encrypted kernel images?
395. How do you configure AppArmor with a custom network and filesystem policy?
396. What steps are required to implement a kernel module for privilege escalation detection?
397. How do you configure SELinux with a custom policy for container isolation?
398. What is the process to implement a secure key exchange with hardware RNG and ECDH?
399. How do you configure nftables for advanced packet filtering with DPI?
400. What steps are needed to implement runtime integrity checking with IMA and EVM?
401. How do you configure IMA with a custom appraisal policy for runtime verification?
402. What is the process to implement secure remote management with SSH and TOTP?
403. How do you configure TOMOYO with a custom domain-based policy for IoT?
404. What steps are required to implement secure IPC in a kernel module with encryption?
405. How do you configure seccomp for a kernel module with syscall restrictions?
406. What is the process to implement secure data storage with hardware-backed encryption?
407. How do you configure grsecurity with a custom policy for kernel hardening?
408. What steps are needed to implement a kernel module for exploit mitigation?
409. How do you configure LUKS with a custom keyslot for multi-user access?
410. What is the process to implement a secure boot chain with U-Boot and verified boot?
411. How do you configure AppArmor with a custom policy for a containerized application?
412. What steps are required to implement a secure logging system with cryptographic signing?
413. How do you configure SELinux for a high-security IoT gateway?
414. What is the process to implement a secure boot with verified initramfs and TPM?
415. How do you configure nftables for stateful packet inspection with custom rules?

416. What steps are needed to implement a kernel module for secure device authentication?
417. How do you configure IMA/EVM for runtime file integrity with custom hashes?
418. What is the process to implement a secure communication protocol with post-quantum crypto?
419. How do you configure seccomp for a high-performance server with minimal syscalls?
420. What steps are required to implement a secure OTA update with digital signatures and anti-rollback?
421. How do you configure a device for secure remote attestation with TPM and IMA?
422. What is the process to implement a secure boot with a custom chain of trust?
423. How do you configure AppArmor for a multi-tenant IoT application?
424. What steps are needed to implement a kernel module for secure memory isolation?
425. How do you configure SELinux with a custom policy for real-time applications?
426. What is the process to implement a secure key storage with a hardware security module?
427. How do you configure nftables for advanced NAT with custom masquerading?
428. What steps are required to implement a secure logging system with remote attestation?
429. How do you configure IMA for runtime integrity with custom policy enforcement?
430. What is the process to implement a secure boot with encrypted device tree blobs?
431. How do you configure TOMOYO for a custom security policy with runtime updates?
432. What steps are needed to implement a kernel module for secure process monitoring?
433. How do you configure seccomp for a containerized application with minimal syscalls?
434. What is the process to implement a secure communication channel with hardware-backed TLS?
435. How do you configure grsecurity for advanced kernel exploit mitigation?
436. What steps are required to implement a secure firmware update with rollback protection?
437. How do you configure LUKS for a high-security storage system with key escrow?
438. What is the process to implement a secure boot with a custom TPM-based chain of trust?
439. How do you configure AppArmor for a high-security network service with minimal permissions?
440. What steps are needed to implement a secure logging system with tamper-proof signatures?

Hardware Interfacing and Peripherals (60 Questions)

441. How do you interface a high-speed ADC with a custom DMA driver and zero-copy transfers?
442. What is the process to configure a MIPI CSI-2 camera with 8K and multi-stream support?
443. How do you implement a driver for a high-resolution e-ink display with partial updates?
444. What steps are required to interface a GNSS module with RTK and SBAS support?
445. How do you configure a PCIe endpoint driver with DMA and MSI-X interrupts?
446. What is the process to implement a high-speed UART driver with FIFO and DMA?
447. How do you interface a high-frequency accelerometer with interrupt-driven sampling?
448. What steps are needed to configure a SPI NOR flash with quad-SPI and ECC?
449. How do you implement a driver for a multi-channel audio codec with 192kHz sampling?
450. What is the process to interface a thermal camera with real-time HDR processing?
451. How do you configure an I2C multiplexer with dynamic bus routing and arbitration?
452. What steps are required to implement a motor controller driver with torque and velocity control?
453. How do you interface a high-speed CAN FD bus with custom error correction?
454. What is the process to configure a PWM fan controller with PID and thermal feedback?
455. How do you implement a driver for a multi-touch controller with gesture recognition?
456. What steps are needed to interface a magnetometer with real-time geomagnetic compensation?
457. How do you configure a USB 3.1 device driver with SuperSpeed and Isochronous transfers?
458. What is the process to implement a driver for an FPGA peripheral with AXI4-Stream?
459. How do you interface a high-speed Ethernet switch with VLAN and QoS support?

460. What steps are required to configure a MIPI DSI display with adaptive refresh and HDR?
461. How do you implement a driver for a high-precision RTC with PPS and atomic clock sync?
462. What is the process to interface a LoRa module with custom modulation and FEC?
463. How do you configure an I2S audio driver with multi-stream and 24-bit depth?
464. What steps are needed to implement a multi-port GPIO expander driver with interrupt chaining?
465. How do you interface a proximity sensor with adaptive ranging and noise filtering?
466. What is the process to configure a CANopen driver with custom PDO and SDO mappings?
467. How do you implement a driver for a high-speed SPI controller with priority queuing?
468. What steps are required to interface a pressure sensor with high-precision compensation?
469. How do you configure a USB-C PD controller with fast role swap and PPS?
470. What is the process to implement a high-speed ADC driver with DMA and oversampling?
471. How do you interface a gyroscope with real-time sensor fusion and 9-DOF?
472. What steps are needed to configure a PCIe switch driver with hot-plug and lane bonding?
473. How do you implement a driver for a high-speed UART with FIFO and error correction?
474. What is the process to interface a LIDAR sensor with real-time point cloud processing?
475. How do you configure an I2C-based power monitor with dynamic range and event triggers?
476. What steps are required to implement a high-resolution camera driver with RAW and YUV?
477. How do you interface a gas sensor with multi-gas detection and calibration?
478. What is the process to configure a SPI-based display controller with gamma correction?
479. How do you implement a driver for a high-precision PWM controller with phase alignment?
480. What steps are needed to interface a vibration sensor with real-time FFT analysis?
481. How do you configure a CAN bus controller with extended frames and error injection?
482. What is the process to implement a SPI flash driver with ECC and wear leveling?
483. How do you interface an IMU with 9-DOF fusion and Kalman filtering?
484. What steps are required to configure an I2C temperature sensor with multi-zone and alerts?
485. How do you implement a driver for a high-speed DAC with multi-channel waveform generation?
486. What is the process to interface an ambient light sensor with spectral compensation?
487. How do you configure a USB gadget driver with multi-function composite and high-speed?
488. What steps are needed to implement an Ethernet controller driver with PTP and TSN?
489. How do you interface a barometer with altitude tracking and temperature compensation?
490. What is the process to configure a SPI-based touch controller with haptics and pressure?
491. How do you implement a driver for a high-speed ADC with differential inputs and oversampling?
492. What steps are required to interface a flow sensor with real-time flow rate calculation?
493. How do you configure an I2C-based EEPROM with secure write and anti-tampering?
494. What is the process to implement a high-speed UART driver with DMA and multi-port?
495. How do you interface a strain gauge with real-time signal processing and calibration?
496. What steps are needed to configure a CAN FD driver with custom error handling and bitrate switching?
497. How do you implement a driver for a high-resolution OLED with adaptive brightness?
498. What is the process to interface a radar sensor with Doppler and range processing?
499. How do you configure a SPI-based motor controller with closed-loop control and PID?
500. What steps are required to implement a high-precision timer driver with event capture?

Notes

- These questions are designed for expert-level candidates with extensive experience in embedded Linux, covering advanced kernel development, device drivers, networking, real-time systems, security, and hardware interfacing.

- Candidates should be proficient with tools like Buildroot, Yocto, gcc, gdb, perf, ftrace, systemtap, and hardware debugging tools (e.g., JTAG, logic analyzers, oscilloscopes).
- Many questions involve custom hardware, requiring access to or simulation with QEMU, FPGA boards, or specific peripherals.
- Practical preparation involves hands-on experience with complex embedded systems, kernel hacking, and real-time optimization.